

RUN Troubleshooting Commands Guide

Essential Windows RUN Commands for Troubleshooting

System Administration & Management

RUN Command	Description	Example
control	Control Panel	control
control admintools	Administrative Tools	control admintools
control printers	Printers and Devices	control printers
control folders	Folder Options	control folders
control keyboard	Keyboard Properties	control keyboard
control mouse	Mouse Properties	control mouse
control userpasswords	User Accounts	control userpasswords2
control netconnections	Network Connections	control netconnections
control schedtasks	Task Scheduler	control schedtasks
control update	Windows Update	control update
compmgmt.msc	Computer Management	compmgmt.msc
devmgmt.msc	Device Manager	devmgmt.msc
diskmgmt.msc	Disk Management	diskmgmt.msc
services.msc	Services	services.msc
gpedit.msc	Group Policy Editor	gpedit.msc
secpol.msc	Local Security Policy	secpol.msc
eventvwr.msc	Event Viewer	eventvwr.msc
perfmon.msc	Performance Monitor	perfmon.msc
taskschd.msc	Task Scheduler	taskschd.msc
fsmgmt.msc	Shared Folders	fsmgmt.msc
dhcpgmt.msc	DHCP Management	dhcpgmt.msc
dnsmgmt.msc	DNS Management	dnsmgmt.msc
dsa.msc	Active Directory Users & Computers	dsa.msc
dsam.msc	Active Directory Sites & Services	dsam.msc
dsa.msc / dssite.msc	AD Services	dsa.msc

Network Troubleshooting

RUN Command	Description	Example
cmd	Command Prompt	cmd
cmd /k	Command Prompt (keep open)	cmd /k ipconfig

RUN Command	Description	Example
powershell	PowerShell	powershell -Command "Get-Service"
powershell -noexit	PowerShell (keep open)	powershell -noexit -Command "Get-Process"
ncpa.cpl	Network Connections	ncpa.cpl
firewall.cpl	Windows Firewall	firewall.cpl
wf.msc	Windows Firewall with Advanced Security	wf.msc
inetcpl.cpl	Internet Properties	inetcpl.cpl
ipconfig	IP Configuration (in cmd)	cmd /k ipconfig /all
ping	Network test (in cmd)	cmd /k ping google.com -t
tracert	Route trace (in cmd)	cmd /k tracert google.com
nslookup	DNS lookup (in cmd)	cmd /k nslookup google.com
netstat	Network stats (in cmd)	cmd /k netstat -ano
pathping	Path ping (in cmd)	cmd /k pathping google.com

System Configuration & Setup

RUN Command	Description	Example
msconfig	System Configuration	msconfig
sysdm.cpl	System Properties	sysdm.cpl
systempropertiesadvanced	Advanced System Settings	systempropertiesadvanced
systempropertiescomputername	Computer Name/Domain	systempropertiescomputername
systempropertieshardware	Hardware Properties	systempropertieshardware
systempropertiesperformance	Performance Options	systempropertiesperformance
systempropertiesdataexecutionprevention	DEP Settings	systempropertiesdataexecutionprevention
systempropertiesremote	Remote Settings	systempropertiesremote
ms-settings:	Windows Settings	ms-settings:system

RUN Command	Description	Example
<code>ms-settings:network</code>	Network Settings	<code>ms-settings:network-wifi</code>
<code>ms-settings:windowsupdate</code>	Windows Update	<code>ms-settings:windowsupdate</code>
<code>ms-settings:privacy</code>	Privacy Settings	<code>ms-settings:privacy-webcam</code>
<code>ms-settings:storage</code>	Storage Settings	<code>ms-settings:storage</code>
<code>ms-settings:apps</code>	Apps Settings	<code>ms-settings:appsfeatures</code>

Disk & Storage Management

RUN Command	Description	Example
<code>diskmgmt.msc</code>	Disk Management	<code>diskmgmt.msc</code>
<code>diskpart</code>	Disk Partition Tool	<code>diskpart</code>
<code>dfrgui</code>	Optimize Drives (Defrag)	<code>dfrgui</code>
<code>cleanmgr</code>	Disk Cleanup	<code>cleanmgr</code>
<code>chkdsk</code>	Check Disk	<code>cmd /k chkdsk C: /f</code>
<code>explorer</code>	File Explorer	<code>explorer C:\</code>
<code>%TEMP%</code>	Temp folder	<code>%TEMP%</code>
<code>%WINDIR%\Temp</code>	System Temp	<code>%WINDIR%\Temp</code>
<code>%USERPROFILE%\Downloads</code>	Downloads folder	<code>%USERPROFILE%\Downloads</code>
<code>%USERPROFILE%\Documents</code>	Documents folder	<code>%USERPROFILE%\Documents</code>
<code>shell:recent</code>	Recent files	<code>shell:recent</code>
<code>shell:Downloads</code>	Downloads	<code>shell:Downloads</code>
<code>shell:Documents</code>	Documents	<code>shell:Documents</code>
<code>shell:Pictures</code>	Pictures	<code>shell:Pictures</code>
<code>shell:Music</code>	Music	<code>shell:Music</code>
<code>shell:Videos</code>	Videos	<code>shell:Videos</code>

User & Profile Management

RUN Command	Description	Example
<code>%USERPROFILE%</code>	User Profile	<code>%USERPROFILE%</code>
<code>%USERNAME%</code>	Username	<code>echo %USERNAME%</code>
<code>netplwiz</code>	User Accounts	<code>netplwiz</code>
<code>control userpasswords</code>	User Passwords	<code>control userpasswords</code>

RUN Command	Description	Example
<code>control userpasswords2</code>	Advanced User Accounts	<code>control userpasswords2</code>
<code>lusrmgr.msc</code>	Local Users & Groups	<code>lusrmgr.msc</code>
<code>whoami</code>	Current user info	<code>whoami /all</code>
<code>rundll32 netplwiz.dll,UsersRunDll</code>	User Accounts Dialog	<code>rundll32 netplwiz.dll,UsersRunDll</code>

Hardware & Device Management

RUN Command	Description	Example
<code>devmgmt.msc</code>	Device Manager	<code>devmgmt.msc</code>
<code>verifier</code>	Driver Verifier	<code>verifier</code>
<code>winver</code>	Windows Version	<code>winver</code>
<code>dxdiag</code>	DirectX Diagnostic Tool	<code>dxdiag</code>
<code>msinfo32</code>	System Information	<code>msinfo32</code>
<code>msdt</code>	Microsoft Support Diagnostic Tool	<code>msdt -id WindowsUpdateDiagnostic</code>
<code>mdsched</code>	Windows Memory Diagnostic	<code>mdsched</code>
<code>resmon</code>	Resource Monitor	<code>resmon</code>
<code>taskmgr</code>	Task Manager	<code>taskmgr</code>
<code>perfmon</code>	Performance Monitor	<code>perfmon</code>
<code>sysmon</code>	System Monitor (if installed)	<code>sysmon -c</code>

Registry & System Configuration

RUN Command	Description	Example
<code>regedit</code>	Registry Editor	<code>regedit</code>
<code>regedt32</code>	Registry Editor (32-bit)	<code>regedt32</code>
<code>gpedit.msc</code>	Group Policy Editor	<code>gpedit.msc</code>
<code>secpol.msc</code>	Local Security Policy	<code>secpol.msc</code>
<code>rsop.msc</code>	Resultant Set of Policy	<code>rsop.msc</code>
<code>gpupdate /force</code>	Update Group Policy	<code>cmd /k gpupdate /force</code>
<code>gpresult /r</code>	Group Policy Results	<code>cmd /k gpresult /r</code>
<code>systeminfo</code>	System Information	<code>cmd /k systeminfo</code>
<code>bcdedit</code>	Boot Configuration	<code>cmd /k bcdedit /enum</code>

Windows Features & Components

RUN Command	Description	Example
appwiz.cpl	Programs and Features	appwiz.cpl
optionalfeatures	Windows Features	optionalfeatures
control /name Microsoft.ProgramsAndFeatures	Programs & Features	control /name Microsoft.ProgramsAndFeatures
appinstaller	App Installer	appinstaller
shell:AppsFolder	All Apps	shell:AppsFolder
scoop	Windows Store App Manager	(if installed)

Internet & Browser

RUN Command	Description	Example
inetcpl.cpl	Internet Options	inetcpl.cpl
iexplore	Internet Explorer	iexplore
chrome	Google Chrome (if installed)	chrome
firefox	Firefox (if installed)	firefox
msedge	Microsoft Edge	msedge
certlm.msc	Certificate Manager (local machine)	certlm.msc
certmgr.msc	Certificate Manager (current user)	certmgr.msc
cryptext	Cryptographic Shell Extension	cryptext

Backup, Recovery & Troubleshooting

RUN Command	Description	Example
sdclt	Backup and Restore	sdclt
systemreset	Reset this PC	systemreset
recoverydrive	Recovery Drive	recoverydrive
rstrui	System Restore	rstrui
sfc /scannow	System File Checker	cmd /k sfc /scannow
DISM /Online /Cleanup-Image /RestoreHealth	DISM Repair	cmd /k DISM /Online /Cleanup-Image /RestoreHealth
chkdsk /f	Check Disk (fix)	cmd /k chkdsk C: /f /r
bootrec	Boot Recovery	cmd /k bootrec /rebuildbcd
msdt	MS Support Diagnostic Tool	msdt /id WindowsUpdateDiagnostic

RUN Command	Description	Example
<code>verifier</code>	Driver Verifier Manager	<code>verifier /standard</code>
<code>sigverif</code>	File Signature Verification	<code>sigverif</code>

Remote & Administration

RUN Command	Description	Example
<code>mstsc</code>	Remote Desktop Connection	<code>mstsc</code>
<code>mstsc /v:servername</code>	Remote to server	<code>mstsc /v:server01</code>
<code>mstsc /admin</code>	Remote to console session	<code>mstsc /admin</code>
<code>tsadmin.msc</code>	Remote Desktop Services Manager	<code>tsadmin.msc</code>
<code>mmc</code>	Microsoft Management Console	<code>mmc</code>
<code>mmc.exe /a</code>	MMC in author mode	<code>mmc.exe /a</code>
<code>compmgmt.msc /computer=servername</code>	Remote Computer Management	<code>compmgmt.msc /computer=server01</code>
<code>telnet</code>	Telnet Client	<code>telnet</code>
<code>ssh</code>	SSH Client (if installed)	<code>ssh user@host</code>

Keyboard Shortcuts & Quick Access

RUN Command	Description	Example
<code>snippingtool</code>	Snipping Tool	<code>snippingtool</code>
<code>calc</code>	Calculator	<code>calc</code>
<code>notepad</code>	Notepad	<code>notepad</code>
<code>wordpad</code>	WordPad	<code>wordpad</code>
<code>mspaint</code>	Paint	<code>mspaint</code>
<code>write</code>	WordPad (alternative)	<code>write</code>
<code>charmap</code>	Character Map	<code>charmap</code>
<code>osk</code>	On-Screen Keyboard	<code>osk</code>
<code>magnify</code>	Magnifier	<code>magnify</code>
<code>narrator</code>	Narrator	<code>narrator</code>
<code>psr</code>	Problem Steps Recorder	<code>psr</code>
<code>sticky notes</code>	Sticky Notes	<code>stikynot</code>
<code>snippingtool /clip</code>	Snipping Tool (clip mode)	<code>snippingtool /clip</code>

Print & Fax

RUN Command	Description	Example
control printers	Printers and Faxes	control printers
printmanagement.msc	Print Management	printmanagement.msc
rundll32 printui.dll,PrintUIEntry	Printer UI	rundll32 printui.dll,PrintUIEntry /?
printui	Printer UI	printui /s /t2

Server & Enterprise Tools

RUN Command	Description	Example
dsa.msc	AD Users & Computers	dsa.msc
dssite.msc	AD Sites & Services	dssite.msc
dsadmin.msc	AD Administrative Center	dsadmin.msc
dhcpgmt.msc	DHCP Management	dhcpgmt.msc
dnsmgmt.msc	DNS Management	dnsmgmt.msc
adsiedit.msc	ADSI Edit	adsiedit.msc
ldp	LDAP Admin Tool	ldp
ntdsutil	NTDS Utility	ntdsutil
repadmin	Replication Admin	cmd /k repadmin /summary
dcdiag	Domain Controller Diagnostics	cmd /k dcdiag /s:dc01
netdom	Domain Management	cmd /k netdom query fsmo
dfsrmig	DFSR Migration	cmd /k dfsrmig /getglobalstate
dfsrmgmt.msc	DFS Management	dfsrmgmt.msc
iskall	IIS Manager (if installed)	iskall
iis.msc	IIS Management Console	iis.msc

Quick Troubleshooting Sequence

cmd

```
:: Step 1 - Basic system info  
winver  
msinfo32
```

```
:: Step 2 - Network check  
cmd /k ipconfig /all  
cmd /k ping 8.8.8.8 -n 4
```

```
:: Step 3 - Service health  
services.msc  
cmd /k sc query state= running
```

```

:: Step 4 - Disk check
cleanmgr
cmd /k chkdsk C: /scan

:: Step 5 - System files
cmd /k sfc /scannow

:: Step 6 - Event logs
eventvwr.msc

:: Step 7 - Performance
perfmon.msc
resmon

:: Step 8 - Update check
control update

```

Common RUN Command Combinations

Combination	Purpose	Example
cmd /k command	Run command and keep window	cmd /k ipconfig /all
powershell -noexit	Keep PowerShell open	powershell -noexit -Command "Get-Process"
runas /user:domain\user	Run as different user	runas /user:ADMIN\admin "cmd /k whoami"
rundll32	Run DLL functions	rundll32 shell32.dll,Control_RunDLL desk.cpl
start	Start in new window	start notepad.exe
explorer	Open file explorer	explorer shell:Downloads

Environment Variables Quick Access

Variable	Path	Description
%USERPROFILE%	C:\Users\username	User home
%APPDATA%	C:\Users\username\AppData\Roaming	Roaming AppData
%LOCALAPPDATA%	C:\Users\username\AppData\Local	Local AppData
%PROGRAMFILES%	C:\Program Files	Program Files
%PROGRAMFILES(X86)%	C:\Program Files (x86)	32-bit Programs
%COMMONPROGRAMFILES%	Common Program Files	Shared files
%SYSTEMROOT%	C:\Windows	Windows folder
%SYSTEMDRIVE%	C:	System drive
%HOMEDRIVE%	C:	Home drive
%HOMEPATH%	\Users\username	Home path
%ALLUSERSPROFILE%	C:\ProgramData	All users data

Variable	Path	Description
%PUBLIC%	C:\Users\Public	Public folder
%TEMP%	C:\Users\username\AppData\Local\Temp	Temp folder

Special Shell Commands

Command	Description	Example
shell:startup	Startup folder	shell:startup
shell:sendto	Send To folder	shell:sendto
shell:desktop	Desktop folder	shell:desktop
shell:favorites	Favorites	shell:favorites
shell:downloads	Downloads	shell:downloads
shell:documents	Documents	shell:documents
shell:pictures	Pictures	shell:pictures
shell:music	Music	shell:music
shell:videos	Videos	shell:videos
shell:recent	Recent items	shell:recent
shell:common desktop	Public Desktop	shell:common desktop
shell:common documents	Public Documents	shell:common documents
shell:common pictures	Public Pictures	shell:common pictures
shell:common music	Public Music	shell:common music
shell:common videos	Public Videos	shell:common videos
shell:common startup	Common Startup	shell:common startup
shell:common downloads	Common Downloads	shell:common downloads
shell:My Pictures	My Pictures	shell:My Pictures
shell:My Music	My Music	shell:My Music
shell:My Video	My Videos	shell:My Video

Troubleshooting Wizards

Command	Description	Example
msdt /id NetworkDiagnostic	Network Troubleshooter	msdt /id NetworkDiagnostic
msdt /id WindowsUpdateDiagnostic	Windows Update Troubleshooter	msdt /id WindowsUpdateDiagnostic
msdt /id AudioPlaybackDiagnostic	Audio Troubleshooter	msdt /id AudioPlaybackDiagnostic
msdt /id PerformanceDiagnostic	Performance Troubleshooter	msdt /id PerformanceDiagnostic

Command	Description	Example
<code>msdt /id PrinterDiagnostic</code>	Printer Troubleshooter	<code>msdt /id PrinterDiagnostic</code>
<code>msdt /id PowerDiagnostic</code>	Power Troubleshooter	<code>msdt /id PowerDiagnostic</code>
<code>msdt /id BluetoothDiagnostic</code>	Bluetooth Troubleshooter	<code>msdt /id BluetoothDiagnostic</code>
<code>msdt /id IEDiagnostic</code>	Internet Explorer Troubleshooter	<code>msdt /id IEDiagnostic</code>
<code>msdt /id DeviceInstallationDiagnostic</code>	Device Installation Troubleshooter	<code>msdt /id DeviceInstallationDiagnostic</code>

Registry Quick Paths

Registry Path	Access via RUN	Description
<code>HKEY_LOCAL_MACHINE\SOFTWARE</code>	<code>regedit</code>	HKLM Software
<code>HKEY_CURRENT_USER\Software</code>	<code>regedit</code>	HKCU Software
<code>HKEY_CURRENT_USER\Control Panel\Desktop</code>	<code>regedit</code>	Desktop settings
<code>HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services</code>	<code>regedit</code>	Services configuration
<code>HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run</code>	<code>regedit</code>	Startup programs
<code>HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run</code>	<code>regedit</code>	Machine startup
<code>HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon</code>	<code>regedit</code>	Winlogon settings
<code>HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer</code>	<code>regedit</code>	Explorer settings
<code>HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Session Manager\Environment</code>	<code>regedit</code>	System environment

Quick Action Commands

cmd

:: --- Show all RUN commands ---
:: Not a direct command, but useful collection

:: Most common troubleshooting sequence

winver
msinfo32
services.msc
devmgmt.msc
diskmgmt.msc
eventvwr.msc
compmgmt.msc

:: Network troubleshooting sequence

cmd /k ipconfig /all
cmd /k ping 8.8.8.8 -t
ncpa.cpl
firewall.cpl
cmd /k netstat -ano

:: System cleanup

cleanmgr
dfrgui
%TEMP%

:: User management

netplwiz
lusrmgr.msc
control userpasswords2

:: Group Policy

gpedit.msc
secpol.msc
cmd /k gpupdate /force
cmd /k gpresult /r